

DAY 3 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time Prediction Using Machine Learning

Day 3: Research Framing & Mentor Review

1. Progress, Blockers & Next Plan

Completed	Current Blockers	Next Plan
Problem understanding	Final dataset not yet inspected	Receive dataset
Initial methodology	Exact columns/labels not confirmed	Inspect structure & metadata
Literature review	Actual data quality cannot yet be assessed	Check missing values/duplicates
Candidate methods		Begin EDA & preprocessing
Expected dataset structure		
Risk identification		

2. Research Question

How accurately can machine learning models estimate battery SoC and predict remaining charging time using historical charging measurements such as voltage, current, power, temperature and charging history?

Sub Questions

- Which model performs best for SoC estimation?
- Which model performs best for remaining charging-time prediction?
- Which features contribute most to prediction?
- How does performance vary across SoC ranges and charging phases?

3. Measurable Outputs

Target 1 — Battery SoC:

Predicted percentage of battery charge.

Target 2 — Remaining Charging Time:

Predicted time left for full charging.

Evaluation Metrics:

MAE

RMSE

R^2

Additional Analysis:

- Feature importance
- Worst predictions
- Error by SoC range
- Error by charging phase